Barathkrishna Satheeshkumar

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Robotics Research Engineering | Robot Learning, Perception and Control

RESEARCH AND ENGINEERING PROJECTS

Autonomous PCB Recycling Robot | Apple-Sponsored CMU Research

- Co-designed robotic recycling system, selected sensors and actuators, designed pneumatic capacitor extraction mechanism
- Achieved sub-millimeter component localization, ran multi-GPU model training and built a real-time vision pipeline with YOLO-OBB, OCR, and stereo PCB tracking through occlusions
- Integrated the UR5e, sensors, and extraction hardware with ROS 2 and MoveIt. Developed motion planning and control for autonomous extraction
- Tested the integrated system and demonstrated autonomous capacitor extraction and collection with duty cycle under 1s

Quadruped Jumping with Reinforcement Learning and Adaptive MPC | CMU

- Integrated adaptive model predictive control (MPC) into the low-level loop of a CAJun-based quadruped jumping system
- Trained a reinforcement learning (RL) policy for centroidal motion targets with the adaptive controller in the simulation loop to study robustness to payload changes, achieving 8x increase in jumping distance due to improved sim-to-real transfer

Action-Informed Estimation and Planning | ICRA 2026

- Developed a mode-based proprioceptive contact estimation algorithm for Boston Dynamics Spot during visual occlusion
- Achieved 85% closed-loop task success versus a 35% open-loop baseline, enabled by the action-conditioned replanning

Flow-CS-PIBT | Multi-Agent Pathfinding

- Developed a 16.7M-parameter rectified-flow GNN that guided collision-free planning for up to 1,000 agents
- Trained the policy on 1.2M samples derived from 3,359 expert trajectories using imitation learning
- Benchmarked on 8 maps: 64.6% all-agent success versus 60.5% for PIBT and 66.4% for the learned baseline

Team Captain and Engineer | IIT Bombay Racing

- Captained a 110+ member Formula Student team and placed 8th overall at Formula Student UK 2022
- Redesigned the planetary gearbox employing CAD, FEA and CFD techniques, reducing gearbox weight by 24%
- Optimized the vehicle dynamics simulation model to run 75% faster, shortening turnaround for tuning and evaluations

PROFESSIONAL EXPERIENCE

Tesla | Software Integration Engineer, Vehicle Firmware

Palo Alto, CA | Jun 2026–Present

- Designed and deployed an ASIL-D longitudinal acceleration safety monitor with active intervention, integrating real-time sensing, filtering, and fault detection on a safety processor; built a displacement monitor to prevent vehicle creep and rollback
- Validated behavior with unit and system tests, SIL tests, and offline replay spanning 10M+ driving hours, in-car validation
- Created test and validation plans for the backup longitudinal controller for Tesla vehicles, aligning with ISO 26262 and SOTIF

PlusAI | Systems Engineering Intern, Autonomous Trucks

Santa Clara, CA | May–Aug 2025

- Built a scenario difficulty scoring tool that achieved 81% accuracy and 78% recall across 144 simulated scenarios
- Improved coverage of replay simulation regression set by over 80% via manual review of vehicle replays and trace analysis

Jaguar Land Rover | Graduate Vehicle Dynamics Engineer

Bengaluru, India | Jul 2022–Jul 2024

- Automated Simulink-to-CarMaker integration and deployed a roll stability simulation tool, cutting runtime from 90 to 3 mins

EDUCATION

Carnegie Mellon University | M.S. Mechanical Engineering (Research) | May 2026

Indian Institute of Technology Bombay | B.Tech. Mechanical Engineering | May 2022

TECHNICAL SKILLS

Programming and ML Python, C++, PyTorch, PyTorch Geometric, reinforcement learning, rectified flow

Robotics and simulation ROS 2, MoveIt, adaptive MPC, motion planning, proprioceptive estimation, Isaac Gym, MuJoCo

Perception and integration YOLO-OBB, OCR, real-time sensing, embedded software, Git, CMake